

R1

<u>sid</u>	<u>bid</u>	<u>day</u>
22	101	10/10/96
58	103	11/12/96

B1

<u>bid</u>	<u>bname</u>	<u>color</u>
101	tiger	red
103	lion	green
105	hero	blue

S1

<u>sid</u>	<u>sname</u>	<u>rating</u>	<u>age</u>
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S2

<u>sid</u>	<u>sname</u>	<u>rating</u>	<u>age</u>
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

Write SQL statements to solve these problems:

1. How to find name and id of sailors who reserved boats in November?
2. How to find name and age of sailors who reserved boats, but not include those who reserved red boat?
3. How to find names of sailors who've reserved all boats?
4. How to find all information of sailors who have reserved more than one boat in one day?
5. How to find names of sailors who have reserved boat #101 and reserved only one time?

1.How to find name and id of sailors who reserved boats in November?

```
SELECT S.sname, S.sid
FROM Sailors AS S, Reserves AS R
WHERE S.sid = R.sid and Month(day)=11;
```

2.How to find name and age of sailors who reserved boats, but not include those who reserved red boat?

```
SELECT S.sname, S.age
FROM Sailors AS S
WHERE S.sid in
  (SELECT sid
   FROM Reserves
  ) AND S.sid not in
  (SELECT sid
   FROM Reserves AS R, Boats AS B
  WHERE R.bid= B.bid AND B.color = 'red ');
```

3.How to find names of sailors who've reserved all boats?

```
SELECT sname
FROM Sailors
WHERE NOT EXISTS
  (SELECT *
   FROM Boats
  WHERE NOT EXISTS
    (SELECT *
     FROM Reserves
    WHERE Sailors.sid = Reserves.sid AND Boats.bid = Reserves.bid));
```

4.How to find all information of sailors who have reserved more than one boat in one day?

```
SELECT *
FROM Sailors AS S
WHERE EXISTS
  (SELECT *
   FROM Reserves AS R1
  WHERE R1.sid= S.sid
  AND EXISTS
    (SELECT *
     FROM Reserves AS R2
    WHERE R2.sid= S.sid AND R2.bid <> R1.bid AND R2.day = R1.day));
```

5.How to find names of sailors who have reserved boat #101 and reserved only one time?

```
SELECT S.sname
FROM Sailors AS S
WHERE EXISTS
(SELECT *
FROM Reserves R1
WHERE R1.bid = 101 AND R1.sid = S.sid AND
NOT EXISTS
(SELECT *
FROM Reserves R2
WHERE R2.bid = 101 AND R2.sid = R1.sid AND R1.day <> R2.day));
```